

TASK 5 CHECKLIST

For both Regression and Classification models

Legend: [BOTH] = do for every model [CLASS] = classification only [REG] = regression only

1. Model Evaluation

- ☐ Calculate: Accuracy, Precision, Recall, F1-score [CLASS]
- ☐ Calculate: RSS, RMSE, R^2 [REG]

2. Check Overfitting / Underfitting

- ☐ Compare Train score vs Test score [BOTH]
- ☐ Train score much higher than Test score → Overfitting
- ☐ Both scores low → Underfitting
- ☐ Train and Test scores close → Good fit 

3. Cross-Validation (5-Fold) OR Bootstrap

- ☐ Run 5-fold (or 10-fold) cross-validation on the training data [BOTH]
- ☐ Note the average score across folds
- ☐ Note the score spread (high spread = unstable model)

4. Compare All Models

- ☐ Fill in one table below (Classification OR Regression) for every model tried

Classification models:

Model	Accuracy	Precision	Recall	F1-score

Regression models:

Model	RSS	RMSE	R^2

- ☐ Pick the model with best score AND stable cross-validation result [BOTH]

5. Hyperparameter Tuning

- ☐ Use **GridSearchCV** or **RandomizedSearchCV** on your best model [BOTH]

- ☐ Try different values of key settings (e.g. `n_estimators`, `max_depth`, `learning_rate`)
- ☐ Note the best combination it finds
- ☐ Re-test the tuned model on the test set — confirm score improved **[BOTH]**

6. Try Advanced Models

- ☐ Random Forest(Bagging) **[BOTH]**
- ☐ AdaBoost **[BOTH]**
- ☐ Gradient Boosting **[BOTH] (Optional)**